

20.4 Cook's distance, DFFITS and DFBETAS

An outlier is a case the model fits badly; an **influential** case is one whose removal would change the conclusions. A remote case that follows the trend of the others exactly has no influence, and a large residual near the centroid moves the fit little. Influence measures, all from Theorem 20.3.1, quantify different consequences of deleting a case.

20.4.1 Four measures

Definition 20.4.1. Cook's distance, DFFITS, DFBETAS

Let $h_{ii} < 1$, and let c_{jj} be the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$, so that $s^2 c_{jj}$ is the estimated variance of $\hat{\beta}_j$.

(a) **Cook's distance** is

$$D_i = \frac{(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})}{p s^2}.$$

(b) **DFFITS** is the change in case i 's own fitted value in units of its standard error,

$$\text{DFFITS}_i = \frac{\hat{y}_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}}{s_{(i)} \sqrt{h_{ii}}}.$$

(c) **DFBETAS** is the change in one coefficient in units of its standard error,

$$\text{DFBETAS}_{ij} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{s_{(i)} \sqrt{c_{jj}}}.$$

(d) **COVRATIO** compares the generalized variances of the estimates with and without case i ,

$$\text{COVRATIO}_i = \frac{\det(s_{(i)}^2 (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1})}{\det(s^2 (\mathbf{X}^\top \mathbf{X})^{-1})}.$$

Cook (1977) proposed D_i ; DFFITS, DFBETAS and COVRATIO are from Belsley, Kuh and Welsch (1980). DFFITS and DFBETAS use $s_{(i)}$, for the same reason as t_i . None requires a refit.

Theorem 20.4.1. Influence identities

Let $h_{ii} < 1$ and $n - p \geq 2$, and let $\tilde{\mathbf{x}}_j$ and ℓ_{ij} be as in Definition 20.2.1.

$$(a) \quad D_i = \frac{\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)}\|^2}{p s^2} = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}.$$

$$(b) \quad \text{DFFITS}_i = t_i \sqrt{h_{ii}/(1 - h_{ii})}, \text{ and } D_i = \text{DFFITS}_i^2 s_{(i)}^2 / (p s^2).$$

$$(c) \quad \text{DFBETAS}_{ij} = t_i \frac{\tilde{x}_{ij}}{\|\tilde{\mathbf{x}}_j\| \sqrt{1 - h_{ii}}}, \text{ so } \text{DFBETAS}_{ij}^2 = t_i^2 \ell_{ij} / (1 - h_{ii}).$$

(d) For every $\boldsymbol{\lambda} \neq \mathbf{0}$,

$$\frac{(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{(i)})^2}{s_{(i)}^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}} \leq \text{DFFITS}_i^2,$$

with equality at $\boldsymbol{\lambda} = \mathbf{x}_{(i)}$. In particular $|\text{DFBETAS}_{ij}| \leq |\text{DFFITS}_i|$ for every j .

(e) $\text{COVRATIO}_i = \frac{1}{1 - h_{ii}} \left(\frac{n - p}{n - p - 1 + t_i^2} \right)^p.$

Proof. Put $\mathbf{d} = \hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} g$ with $g = \hat{\varepsilon}_i / (1 - h_{ii})$ (Theorem 20.3.1(b)).

(a) $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)} = \mathbf{X} \mathbf{d}$, and $\|\mathbf{X} \mathbf{d}\|^2 = \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d} = g^2 \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = g^2 h_{ii}$. So $p D_i s^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2$, and dividing by s^2 and using $r_i^2 = \hat{\varepsilon}_i^2 / \{s^2(1 - h_{ii})\}$ gives (a).

(b) The numerator of DFFITS_i is $\mathbf{x}_{(i)}^\top \mathbf{d} = h_{ii} g$, so $\text{DFFITS}_i = \sqrt{h_{ii}} \hat{\varepsilon}_i / \{s_{(i)}(1 - h_{ii})\} = t_i \sqrt{h_{ii} / (1 - h_{ii})}$. Then $\text{DFFITS}_i^2 s_{(i)}^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2 = p D_i s^2$.

(c) By Theorem 6.6.2(a) the j th row of $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$, so the j th entry of $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$, the i th column of that matrix, is $\tilde{x}_{ij} / \|\tilde{\mathbf{x}}_j\|^2$. By Eq. 6.6.1, $c_{jj} = 1 / \|\tilde{\mathbf{x}}_j\|^2$. Hence

$$\text{DFBETAS}_{ij} = \frac{\tilde{x}_{ij} g \|\tilde{\mathbf{x}}_j\|^{-2}}{s_{(i)} \|\tilde{\mathbf{x}}_j\|^{-1}} = \frac{\tilde{x}_{ij} \hat{\varepsilon}_i}{s_{(i)} \|\tilde{\mathbf{x}}_j\| (1 - h_{ii})},$$

which is the stated form.

(d) $\boldsymbol{\lambda}^\top \mathbf{d} = g \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$. Let $\mathbf{K}^{1/2}$ be the symmetric square root of $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 1.7.6). The Cauchy-Schwarz inequality (Proposition 1.1.1) for $\mathbf{K}^{1/2} \boldsymbol{\lambda}$ and $\mathbf{K}^{1/2} \mathbf{x}_{(i)}$ gives $(\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)})^2 \leq \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \cdot h_{ii}$, with equality when $\boldsymbol{\lambda}$ is a multiple of $\mathbf{x}_{(i)}$. So the ratio is at most $g^2 h_{ii} / s_{(i)}^2 = \text{DFFITS}_i^2$. Taking $\boldsymbol{\lambda} = \mathbf{e}_j$ gives DFBETAS_{ij}^2 .

(e) By Example (Deleting an observation), $\det(\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)}) = (1 - h_{ii}) \det(\mathbf{X}^\top \mathbf{X})$, so $\text{COVRATIO}_i = (s_{(i)}^2 / s^2)^p / (1 - h_{ii})$. From the proof of Theorem 20.3.3(c), $s_{(i)}^2 / s^2 = (n - p - r_i^2) / (n - p - 1)$, and solving $t_i^2 = r_i^2 (n - p - 1) / (n - p - r_i^2)$ for r_i^2 gives $n - p - r_i^2 = (n - p)(n - p - 1) / (n - p - 1 + t_i^2)$. Substitute.

■

20.4.2 What the measures mean

Cook's distance has two readings. By (a) it is the squared distance the fitted vector moves when case i is deleted, in units of $p s^2$. By Theorem 12.2.1 with $\boldsymbol{\Lambda} = \mathbf{I}$, the $100(1 - \alpha)\%$ confidence ellipsoid for $\boldsymbol{\beta}$ is $\{\mathbf{b} : (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}}) \leq p s^2 F_\alpha(p, n - p)\}$, so $\hat{\boldsymbol{\beta}}_{(i)}$ lies on the boundary of the ellipsoid of level $\Pr\{F(p, n - p) \leq D_i\}$.

By the second equality in (a), D_i is misfit, r_i^2 , times leverage, $h_{ii} / (1 - h_{ii})$: a case is influential only if it has both. DFFITS factors the same way with t_i . The **influence plot** of Figure 20.4.1 places each case by leverage and studentized residual, with the curves of constant D_i , $r^2 = p D(1 - h) / h$.

DFBETAS isolates one coefficient: by (c) the partial leverage ℓ_{ij} replaces the leverage in the product, but the factor $1 / (1 - h_{ii})$ still involves the overall leverage. By (d), the largest standardized change in *any* linear combination $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is $|\text{DFFITS}_i|$, so a small DFFITS rules out

a large change in every coefficient and contrast. **COVRATIO** measures precision rather than location: by (e) it exceeds one when h_{ii} is large and t_i small (the case adds precision) and falls below one when t_i is large (the case inflates s^2).

20.4.3 The null law of Cook's distance and the choice of cut-offs

How large must an influence measure be before it matters? Since D_i is a multiple of r_i^2 , its law under the model is known exactly.

Proposition 20.4.2. Cook's distance under the model

Assume Eq. 7.3.1 with $n - p \geq 2$ and all leverages below one. Then $D_i = \frac{n-p}{p} \frac{h_{ii}}{1-h_{ii}} B_i$ with $B_i \sim \text{Beta}(\frac{1}{2}, \frac{n-p-1}{2})$, so

$$E(D_i) = \frac{h_{ii}}{p(1-h_{ii})}, \quad \frac{1}{n} \sum_{i=1}^n E(D_i) \geq \frac{1}{n-p},$$

with equality in the last bound iff all leverages equal p/n .

Proof. Combine [Theorem 20.4.1\(a\)](#) with [Proposition 20.1.1\(a\)](#), and use $E(B_i) = 1/(n-p)$. For the bound, $u \mapsto u/(1-u)$ is strictly convex on $[0, 1)$, so by Jensen's inequality the average of $h_{ii}/(1-h_{ii})$ is at least $\bar{h}/(1-\bar{h}) = p/(n-p)$, with equality iff the h_{ii} are equal. ■

So under the model a typical D_i is of order $1/n$, which explains the popular cut-off $4/n$. For the state design of [Example \(How much does the District of Columbia matter?\)](#) the average of $E(D_i)$ is 0.0240, against the bound $1/(n-p) = 0.0213$. Comparing D_i with the median of $F(p, n-p)$, which is near one, is far stricter: it flags only cases whose deletion moves the estimate to the edge of a 50% confidence region. The usual cut-offs are these.

Measure	Cut-off	Reasoning
h_{ii}	$2p/n$	twice the average leverage
D_i	$4/n$, or the median of $F(p, n-p)$	four times the typical value; or the edge of the 50% region
$ DFFITS_i $	$2\sqrt{p/n}$	$ t_i \approx 2$ at a case of average leverage p/n
$ DFBETAS_{ij} $	$2/\sqrt{n}$	$ t_i \approx 2$ at a case of average partial leverage $1/n$
$ \text{COVRATIO}_i - 1 $	$3p/n$	first-order expansion of (e) at $ t_i \approx 2$ or $h_{ii} \approx 2p/n$

All are arbitrary in the same way: each puts a case with studentized residual about two and typical leverage near the line. Calibrating D_i exactly by [Proposition 20.4.2](#) would not help: for a fixed design D_i is a multiple of r_i^2 , so the resulting test is an outlier test with a case-dependent

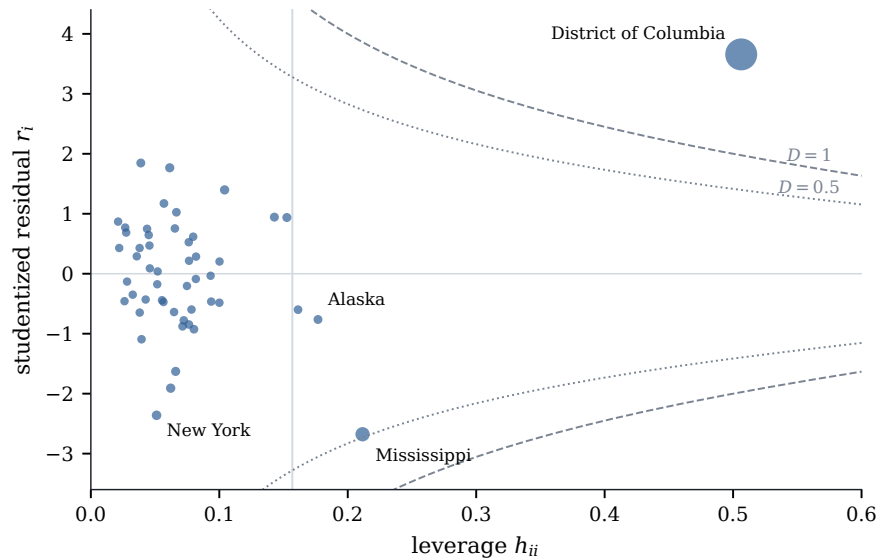


Figure 20.4.1: Influence plot for the state regression: leverage against r_i , area proportional to Cook's distance, curves $D_i = 0.5$ (dotted) and $D_i = 1$ (dashed), and the line $h_{ii} = 2p/n$.

level, discarding the leverage factor that is the point of an influence measure. Influence is a consequence to be assessed, not a hypothesis: the right threshold is whether deleting the case changes a conclusion.

Example. How much does the District of Columbia matter?

In the murder-rate regression with all 51 jurisdictions, the District of Columbia has $D_i = 3.430$, seven times the next largest (Mississippi, 0.481); the median of $F(4, 47)$ is 0.851, and deleting the District moves $\hat{\beta}$ to the boundary of the 0.985 confidence ellipsoid. Its DFFITS is 4.333 (cut-off 0.560) and its DFBETAS for poverty -1.090 (cut-off 0.280). Its partial leverage for poverty is only 0.032 (Example (Where the District of Columbia's leverage comes from)), but by [Theorem 20.4.1\(c\)](#) it is multiplied by $t_i^2/(1 - h_{ii})$: $4.278^2 \times 0.032/(1 - 0.506) \approx 1.19$, so $|\text{DFBETAS}| \approx 1.09$. Alaska, with the largest partial leverage (0.141), has $t_i = -0.760$ and DFBETAS 0.315.

The consequences are substantive. With the District, the poverty coefficient is 0.1551 (standard error 0.1059, $t = 1.47$); without it, 0.2538 (standard error 0.0934, $t = 2.72$), the fit of Example (Poverty and murder rates). One jurisdiction decides whether poverty has a detectable association with murder rates given the other regressors, and the single-parent coefficient moves from 0.6256 to 0.3980. The cut-offs flag 2 cases by $D_i > 4/n = 0.078$, 3 by DFFITS, 5 by DFBETAS for poverty and 4 by leverage above $2p/n = 0.157$, but only the District changes a conclusion. Its COVRATIO is 0.578: it inflates the estimated generalized variance through its contribution to s^2 .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
kept
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
names = np.array(data.index)
n, p = X.shape
```

```

Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta                                # raw residuals
h = np.sum(Q ** 2, axis=1)                        # leverages
s2 = e @ e / (n - p)
r = e / np.sqrt(s2 * (1 - h))                    # internally studentized
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1) # s^2 without case i
t = e / np.sqrt(s2_del * (1 - h))                # externally studentized

cook = r ** 2 * h / (p * (1 - h))                # Cook's distance
dffits = t * np.sqrt(h / (1 - h))
K = np.linalg.inv(X.T @ X)
C = K @ X.T                                     # row j: weights of y in beta_j
dfbetas = (C * e / (1 - h)).T / np.sqrt(np.outer(s2_del, np.diag(K)))
for i in np.argsort(-cook)[:3]:
    print(f"{names[i]:22s} D = {cook[i]:.3f}  DFFITS = {dffits[i]:6.3f}"
          f"  DFBETAS(poverty) = {dfbetas[i, 1]:6.3f}")
print("conventional cut-offs: D > 4/n =", round(4 / n, 3),
      " |DFFITS| > 2 sqrt(p/n) =", round(2 * np.sqrt(p / n), 3),
      " |DFBETAS| > 2/sqrt(n) =", round(2 / np.sqrt(n), 3))

```

[Run this code online](#)

20.4.4 Exercises

20.4.4.1 A. Check your understanding

Exercise. A1

In a regression with $p = 3$, case i has $r_i = 2$. Compute D_i when $h_{ii} = 0.1$ and when $h_{ii} = 0.6$. Which case would you look at first, and why might neither be alarming?

Solution. By Theorem 20.4.1(a), $D_i = (4/3) h_{ii} / (1 - h_{ii})$: $(4/3)(1/9) = 4/27 \approx 0.15$ and $(4/3)(1.5) = 2$. The second moves the coefficient vector to about the edge of a high-level confidence region. Neither is alarming by itself if the change it produces in the estimates of interest is small relative to their uncertainty, or if the case is known to be correct and the model to hold in its region.

Exercise. A2

Show that $D_i = 0$ iff $\hat{\varepsilon}_i = 0$, whatever the leverage. Give an example of a case with leverage 0.9 and no influence.

20.4.4.2 B. Practice

Exercise. B1

Let $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. Show that the sum of squares of the j th row of $\mathbf{C}$ is c_{jj} , and deduce that $\sum_i \ell_{ij} = 1$ in a second way. Why does this suggest a cut-off of order $1/\sqrt{n}$ for DFBETAS?

Solution. $\mathbf{C}\mathbf{C}^\top = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1}$, whose j th diagonal entry is c_{jj} . The j th row of $\mathbf{C}$ is $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$ (proof of [Theorem 20.4.1\(c\)](#)), so $\sum_i \ell_{ij} = \sum_i \tilde{x}_{ij}^2 / \|\tilde{\mathbf{x}}_j\|^2 = c_{jj} \|\tilde{\mathbf{x}}_j\|^2 = 1$. The partial leverages for a coefficient average $1/n$. With $|t_i| \approx 2$ and h_{ii} small, $|\text{DFBETAS}_{ij}| \approx 2\sqrt{\ell_{ij}} \approx 2/\sqrt{n}$ for a typical case.

Exercise. B2

For simple regression, show that the DFBETAS of case i for the slope is $t_i(x_i - \bar{x}) / \sqrt{S_{xx}(1 - h_{ii})}$. Which cases can change the slope most?

Exercise. B3

Use [Theorem 20.4.1\(e\)](#) to show that for large n and small p/n , $\text{COVRATIO}_i \approx 1 + h_{ii} - p(t_i^2 - 1)/n$. Deduce the cut-off $|\text{COVRATIO}_i - 1| > 3p/n$ from the two extreme cases $h_{ii} \approx 0$, $|t_i| = 2$ and $h_{ii} = 2p/n$, $t_i = 0$.

Solution. $1/(1 - h) \approx 1 + h$, and $((n - p)/(n - p - 1 + t^2))^p = (1 + (t^2 - 1)/(n - p))^{-p} \approx 1 - p(t^2 - 1)/n$. Multiplying, $\text{COVRATIO} \approx 1 + h - p(t^2 - 1)/n$ to first order. With $h \approx 0$ and $t^2 = 4$ this is $1 - 3p/n$: a case of no leverage and a residual of two standard errors lowers the precision by that much. With $t = 0$ and $h = 2p/n$ it is $1 + 3p/n$: a well-fitting case of high leverage raises it by the same amount. Deviations beyond $3p/n$ in either direction are more extreme than both.

20.4.4.3 C. Going deeper

Exercise. C1

How influential can one case be, whatever its response? Fix a design of full rank with $h_{ii} < 1$ for all i , and let $\mathbf{y}$ range over $\mathbb{R}^n$.

- Show that $\hat{\varepsilon}_i = \{(\mathbf{I} - \mathbf{M})\mathbf{e}_i\}^\top \hat{\boldsymbol{\varepsilon}}$, and deduce from the Cauchy-Schwarz inequality (Proposition 1.1.1) that $r_i^2 \leq n - p$ for *every* data vector, with equality iff $\hat{\boldsymbol{\varepsilon}}$ is a multiple of $(\mathbf{I} - \mathbf{M})\mathbf{e}_i$. (Proposition 20.1.1(b) gives the same bound from the beta law; this argument needs no model at all.)
- Deduce $D_i \leq \frac{n - p}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}$, and show that the bound is attained: take $\mathbf{y} = \mathbf{e}_i$.
- Conclude that in a design with equal leverages $h_{ii} = p/n$ no case can have $D_i > 1$, whatever the responses. Using the confidence-ellipsoid reading of D_i , say what this means for how far one case can move $\hat{\boldsymbol{\beta}}$.
- Contrast a design with a leverage close to one, and explain how (b) makes the $2p/n$ cut-off for leverage a statement about the *worst case* rather than the observed one.

Solution.

- Since $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$, $\hat{\varepsilon}_i = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \{(\mathbf{I} - \mathbf{M})\mathbf{e}_i\}^\top \{(\mathbf{I} - \mathbf{M})\mathbf{y}\}$. By Cauchy-Schwarz, $\hat{\varepsilon}_i^2 \leq \|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2 \|\hat{\boldsymbol{\varepsilon}}\|^2 = (1 - h_{ii}) \text{SSE}$, with

equality iff the two vectors are proportional. Divide by $s^2(1-h_{ii}) = \text{SSE}(1-h_{ii})/(n-p)$ to get $r_i^2 \leq n-p$.

- (b) Substitute in [Theorem 20.4.1\(a\)](#), $D_i = r_i^2 h_{ii} / \{p(1-h_{ii})\}$. For $\mathbf{y} = \mathbf{e}_i$ the residual is $\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{M})\mathbf{e}_i$, which is proportional to itself, so equality holds in (a) and D_i attains the bound.
- (c) With $h_{ii} = p/n$ the bound is $\frac{n-p}{p} \cdot \frac{p/n}{1-p/n} = 1$. By the ellipsoid reading, deleting any one case moves $\hat{\boldsymbol{\beta}}$ at most to the boundary of the confidence ellipsoid of level $\Pr\{F(p, n-p) \leq 1\}$, which is near 0.5 for moderate $n-p$: in a design that spreads leverage evenly, no single case, however wild its response, can move the estimate beyond a middling confidence region.
- (d) The bound grows without limit as $h_{ii} \rightarrow 1$, and the same case would then also have a tiny residual, so the *observed* D_i need not be large. Leverage is the design's exposure to a bad response: a case with h_{ii} near one can do unbounded damage if its y_i is wrong, and nothing in the data reveals whether it is. That is why leverage is screened on its own, before any residual is looked at.

Exercise. C2

Cook's distance and DFFITS differ only in using s or $s_{(i)}$, but they can rank cases differently. Take $n-p=6$ and two cases with

$$h_{11} = \frac{10}{19}, \quad r_1^2 = 1, \quad h_{22} = \frac{1}{10}, \quad r_2^2 = 5.$$

- (a) Using [Theorem 20.3.3\(c\)](#), show that $t_1^2 = 1$ and $t_2^2 = 25$.
- (b) Show that $pD_1 = 10/9$ and $pD_2 = 5/9$, while $\text{DFFITS}_1^2 = 10/9$ and $\text{DFFITS}_2^2 = 25/9$. So D ranks case 1 first and DFFITS ranks case 2 first.
- (c) Check that such a data set is possible: find p and n for which the constraints $\sum_i h_{ii} = p$ and $\sum_i (1-h_{ii})r_i^2 = n-p$ ([Exercise \(A2\)](#)) can be met by the remaining cases.
- (d) Explain the disagreement in one sentence, and say which measure you would use to decide whether case 2 is an outlier and which to decide whether it matters.

Solution.

- (a) $t_i^2 = r_i^2(n-p-1)/(n-p-r_i^2) = 5r_i^2/(6-r_i^2)$, so $t_1^2 = 5/5 = 1$ and $t_2^2 = 25/1 = 25$. The second case has a residual near the algebraic maximum $r^2 \leq n-p = 6$ of [Exercise \(C1\)](#), so deleting it shrinks s^2 sharply and t_2^2 is much larger than r_2^2 .
- (b) With $g_i = h_{ii}/(1-h_{ii})$ we have $g_1 = (10/19)/(9/19) = 10/9$ and $g_2 = 1/9$. Then $pD_i = r_i^2 g_i$ by [Theorem 20.4.1\(a\)](#), giving 10/9 and 5/9; and $\text{DFFITS}_i^2 = t_i^2 g_i$ by [Theorem 20.4.1\(b\)](#), giving 10/9 and 25/9. So $D_1 = 2D_2$ while $\text{DFFITS}_2^2 = 2.5 \text{DFFITS}_1^2$.
- (c) Take $p = 3$ and $n = 9$. The two cases use $10/19 + 1/10 = 119/190$ of the leverage budget $\sum_i h_{ii} = 3$, leaving $451/190$ for the other seven, each of which can then have leverage well below one. They use $1 \cdot \frac{9}{19} + 5 \cdot \frac{9}{10} = 189/38$ of the budget $\sum_i (1-h_{ii})r_i^2 = n-p = 6$, leaving $39/38$ for the other seven. Both constraints can be met.
- (d) D_i measures the move against s , which the case's own residual inflates, while DFFITS measures it against $s_{(i)}$, which it does not; so a case with a large residual looks worse to DFFITS, and a case with high leverage and a modest residual looks worse to D_i . To ask whether case 2 is an outlier, use t_i , which has an exact null law ([Theorem 20.3.3\(b\)](#));

to ask whether it matters, use D_i , whose ellipsoid reading is a statement about the estimate rather than about the error.